

Chapter 66 — Surgical, Radiographic, and Endoscopic Anatomy of the Male Reproductive System

SCROTUM

Anatomy

Gross Structure

- The testis and spermatic cord are invested in **six tissue layers**, a consequence of **descent from the retroperitoneum**; each layer is a surgical plane in scrotal surgery.
- **Scrotal skin**: hair-bearing, pigmented, rugated; sebaceous follicles and sweat glands.
- **Median raphe** marks the **scrotal septum**, running in the midline from the urethral meatus to the anus and dividing the testes into separate compartments.
- Layer sequence, superficial to deep: **skin → tunica dartos → external spermatic fascia → cremasteric fascia → internal spermatic fascia → tunica vaginalis** (parietal, then visceral).

SCROTAL AND CORD LAYERS — ORIGIN AND FUNCTION

Layer	Continuous with	Content / function
Tunica dartos	Scarpa fascia (abdomen), Colles fascia (perineum), dartos of the penis	Smooth muscle
External spermatic fascia	External oblique fascia	Fascial layer
Cremasteric fascia	Internal oblique	Striated cremaster muscle ; retracts the testis; thermoregulation for spermatogenesis
Internal spermatic fascia	Transversalis fascia	Fascial layer
Parietal tunica vaginalis	Peritoneum	Outer wall of the potential space in which a hydrocele forms
Visceral tunica vaginalis	Peritoneum	Invests the testis; intimately fused with the tunica albuginea

- What the table tests: **each scrotal layer is an abdominal-wall layer carried down** by testicular descent. The trap is the cremaster — **striated** and derived from the **internal oblique**, not the external.
- Tunica vaginalis is continuous with the testis posterolaterally at its **mesentery**, where it attaches to the scrotal wall; the **gubernaculum fixes the inferior pole**.

Arterial Supply

- **Anterior wall**: external pudendal arteries; **posterior wall**: perineal branches.
- Spermatic fasciae: **cremasteric, testicular and deferential** branches.
- Scrotal arteries **run parallel to the rugae and do not cross the median raphe**.

Venous Drainage

- Anterior wall drains via the **external pudendal veins**, parallel to the rugae and **not crossing the raphe**.

Lymphatic Drainage

- Scrotal lymphatics do not cross the raphe and drain to the **ipsilateral superficial inguinal nodes**.

Nerve Supply

ANTERIOR VERSUS POSTERIOR SCROTAL WALL

	Anterior wall	Posterior wall
Artery	External pudendal	Perineal branches
Vein	External pudendal	—
Nerve	Ilioinguinal and genitofemoral branches	Scrotal branches of the perineal nerve; posterior femoral cutaneous nerve

- Anterior nerves also run parallel to the rugae and do not cross the raphe.

KEY POINTS: SCROTUM

- The smooth muscle of the dartos fascia under the scrotal skin is continuous with **Colles fascia, Scarpa fascia and the dartos fascia of the penis**.
- Scrotal wall arteries run **parallel to the rugae** and do not cross the median raphe.
- Branches of the **ilioinguinal and genitofemoral nerves** innervate the anterior scrotal wall.

Concept — *The scrotum is abdominal wall carried into the perineum: every layer has an abdominal counterpart, and nothing — artery, vein, lymphatic or nerve — crosses the median raphe, so each hemiscrotum is a self-contained unit draining to its own groin.*

TESTIS

Anatomy

Gross Structure

- Paired organs with **exocrine and endocrine** function; **volume 15–25 mL, longitudinal axis 4.5–5.1 cm**.
- **Appendix testis**: small pedunculated structure at the **superior pole**; a **müllerian (paramesonephric) duct remnant**, present in some males.
- Capsule, external to internal: **visceral tunica vaginalis → tunica albuginea → tunica vasculosa**.
- **Tunica albuginea**: thick fibroelastic capsule with **smooth muscle cells through collagen** → possible capsular contractility → proposed roles in **regulating testicular blood flow** and **propelling seminiferous tubule fluid out of the testis**.
- Posteriorly the capsule projects inward as the **mediastinum testis** — an **echogenic band parallel to the epididymis** on ultrasonography.
- The epididymis attaches on the **posterolateral** aspect.

Microanatomic Architecture

- Septa radiate from the mediastinum, forming **about 250 wedge-shaped lobules**, each containing **at least one centrifugal artery**.
- **600–1200 seminiferous tubules** per testis, total length **280–400 m**; long, coiled and looped, terminating in the **rete testis**.
- Tubules contain **developing germ cells** and supporting cells: **Sertoli cells, fibrocytes and myoid cells** of the basement membrane.
- **Interstitium = 20–30% of testicular volume**: Leydig cells, mast cells, macrophages, nerves, blood vessels, lymphatics.

LEYDIG VERSUS SERTOLI CELLS

	Leydig cell	Sertoli cell
Compartment	Interstitial — loose tissue between tubules	Tubular — lines the tubule on the basement membrane
Function	Testosterone production	Supports germ cell development
Signature feature	—	Tight junctions divide basal and adluminal compartments → blood–testis barrier

Sperm outflow sequence: seminiferous tubule → **rete testis** → **6–12 ductuli efferentes** (carry fluid and sperm into the caput) → convolute, enlarge and form conical lobules, each with one duct → **single epididymal duct** (about **6 m** uncoiled) → caput, corpus and cauda within a fibrous sheath → duct thickens and straightens → **vas deferens** at the tail.

Figure legends

- Retrograde microbead injection through the rete testis outlines the seminiferous tubules (mouse testis, architecture closely resembling human).

Arterial Supply**THE THREE ARTERIAL SOURCES OF THE TESTIS**

Artery	Origin	Territory	Point to know
Testicular (internal spermatic)	Aorta, below the renal artery	Main supply	Diameter exceeds deferential and cremasteric combined
Deferential (artery of the vas)	Superior or inferior vesical	Vas deferens, cauda epididymis	Collateral; compromised by vasectomy
Cremasteric (external spermatic)	Inferior epigastric	Tunica vaginalis	Anastomoses at the mediastinum

- Course: aorta → **crosses psoas and ureter** → inguinal ring → spermatic cord. **Branching most often occurs within the inguinal canal**; the number of branches entering the testis varies (single most commonly).
- **Countercurrent heat exchange** with the pampiniform plexus → cord arterial blood **2–4 °C below rectal temperature**.
- Near the mediastinum the artery is **highly coiled** and branches; it runs **inferiorly along the posterior surface** within the parenchyma, supplying the **anterior upper pole** and the **anterior, medial and lateral lower pole**.
- **Centrifugal arteries** pass in the septa → arterioles → **intertubular and peritubular capillaries**.
- Testicular–deferential **intercommunications may preserve viability** after testicular artery ligation, yet **atrophy and/or azoospermia** have followed ligation in adults and children.
- Surgical rules: **no traction suture through the lower pole**; **biopsy the medial or lateral surface of the upper pole**; the **midportion carries fewer vessels** than either pole.
- **Prior vasectomy** → deferential artery may be compromised → in later cord or testicular surgery **preserve the testicular artery** and **counsel on devascularisation risk**.

Venous Drainage

- Testicular veins are **anomalous: they do not travel with their arteries**.
- Parenchymal veins drain to a **mediastinal group** or a **surface group**; these coalesce with the deferential veins as the **pampiniform plexus**.

THE THREE VENOUS GROUPS OF THE PAMPINIFORM PLEXUS

Group	Accompanies	Drains to
Anterior (internal spermatic)	Testicular artery	Inferior vena cava (right) / left renal vein (left)
Middle (deferential)	Vas deferens	Pelvic veins
Posterior (external spermatic)	Posterior spermatic cord	Superficial and deep inferior epigastric and superficial and deep pudendal veins

- This arrangement enables **countercurrent heat exchange**; a **varicocele disrupts drainage and raises scrotal temperature**.
- After varicocele correction the **middle and posterior groups provide collateral return**.
- Plexus → **two or three veins in the inguinal canal** → **one vein** ascending to the inferior vena cava (right) or renal vein (left).

RIGHT VERSUS LEFT TESTICULAR VEIN

	Right	Left
Insertion	Inferior vena cava, obliquely	Left renal vein
Consequence	Oblique entry acts as a natural valve against reflux	Anatomic predisposition to varicocele — about 90% of varicoceles are left-sided

- Anastomoses with the **external pudendal, cremasteric and vasal veins** may explain **varicocele recurrence after embolisation**.

VARICOCELE CORRECTION — LEVEL OF LIGATION

	High ligation	Inguinal / subinguinal
Level	Above the exit of deferential vessels and external veins from the cord	Within or below the inguinal canal
Structures ligated	Testicular artery and vein	Veins, sparing the artery
Collaterals	Sufficient to maintain viability	Less extensive
Devascularisation risk	Low	Heightened if the testicular artery is inadvertently ligated

Figure legends

- The pampiniform plexus connects with the **saphenous, internal iliac and external iliac veins**.

Lymphatic Drainage

- Testicular lymph → **para-aortic and interaortocaval nodes**, ascending within the spermatic cord.
- Lymph capillaries **do not penetrate the seminiferous tubules**.
- Lymphatic disruption at varicocele correction → **hydrocele**; the **microsurgical approach preserves lymphatics** and carries a **lower hydrocele risk**.

TESTIS VERSUS SCROTUM — LYMPHATIC DRAINAGE

Testis	Scrotal skin
Retroperitoneal : para-aortic, interaortocaval (follows embryological origin)	Ipsilateral superficial inguinal ; does not cross the raphe

Nerve Supply

- **Visceral (autonomic) innervation** arises in the **renal and aortic plexuses** and runs with the gonadal vessels; the testis has **no known somatic innervation**.
- The **pelvic plexus**, via the vas deferens, adds afferent and efferent fibres.

- Functional role still being defined; postulated roles in **Leydig cell steroidogenesis** and **vascular tone**.
- “**Trifecta complex**” — three cord nerve distributions thought to drive **chronic orchialgia**: **perivasal, posterior periarterial/lipomatous, intracremasteric**. **Microsurgical spermatic cord denervation** targets these.
- Some fibres **cross to the contralateral pelvic plexus** → may explain **unilateral pathology affecting the contralateral testis** (reported with varicocele and testicular tumour).
- **Genital branch of the genitofemoral nerve**: sensation to the **parietal and visceral tunica vaginalis and overlying scrotum**; travels with the testicular artery, ramifies in the tunica albuginea, **never enters the seminiferous tubules**; the seminiferous epithelium is nerve-free.

Blood-Testis Barrier

- **Primary level: Sertoli-Sertoli tight junctions** → basal and adluminal compartments; **premeiotic germ cells separated** from other germ cells.
- Additional levels: **capillary endothelium** and **peritubular myoid cells**.
- Evidence: fluid leaving the tubules and testis differs substantially from plasma and lymph — the **anatomic component**.
- **Functional component** may isolate the haploid gamete or support meiosis; **significance unproven** (further in Chapter 67).

Imaging

Ultrasonography

- **Ultrasonography (US) is the primary modality** for the scrotum and contents: **7.5–10 MHz** transducers, grey-scale real-time, colour flow and power Doppler.
- Indications: scrotal pain, **varicocele recurrence after repair, equivocal examination**, testicular mass, swelling, epididymitis or orchitis.
- In scrotal pain most studies are normal or benign; only a **small minority** show a finding needing urgent surgery.
- Technique: **supine, warm room, warmed gel**.
- Normal findings: scrotal wall **3–4 mm, hypoechoic**; small **anechoic physiological fluid** between the tunica vaginalis layers (more = hydrocele); **mediastinum = posterior echogenic band** parallel to the epididymis; testis **fine, homogeneous, medium-level echoes**, about $5 \times 3 \times 2$ cm; colour Doppler shows vessels in most.
- Waveforms: intratesticular and capsular arteries — **low impedance, high diastolic flow** (low testicular vascular resistance); supratesticular testicular, deferential and cremasteric arteries also low impedance.
- **Suspected torsion → prompt surgical exploration**; Doppler may be used in the acute scrotum but must not delay it.
- Infertility: US **rarely indicated**; used for varicocele and palpable masses. Varicocele study: cord examined **erect with Valsalva**; Doppler shows **persistent reflux after repair**.
- **Varicocele is a clinical diagnosis**; no agreed significant vein size; US reserved for equivocal examination or when examination is limited (**cord lipoma, fatty infiltration**).

Computed Tomography

- Computed tomography (CT) is not primary; useful for **inguinal and cord pathology**; scrotal findings often **incidental**. Shows **air in Fournier gangrene** and hydrocele fluid.

Magnetic Resonance Imaging

- Magnetic resonance imaging (MRI) is a **secondary modality** for **indeterminate or nondiagnostic US**.
- Clearly separates **tunica albuginea, parenchyma and epididymis**; distinguishes **intra- from extratesticular, solid from cystic, complicated from simple** collections; resolves structures **greater than 1 mm**.
- Advantage: **both hemiscrota and the inguinal region in one plane**; also characterises cord pathology.
- **Magnetic resonance angiography** has identified **inguinal testicular nubbin** in the nonpalpable testis.
- Example: nonobstructive azoospermia with **bilateral testicular atrophy** (volumes 5–6 mL, homogeneous signal) and normal cord structures.

SCROTAL IMAGING MODALITIES COMPARED

Modality	Role	Distinctive use
US	First line	Mass, pain, varicocele (erect, Valsalva), Doppler flow
CT	Not primary; often incidental	Fournier gangrene (tissue air), inguinal and cord disease
MRI	Second line	Indeterminate US; intra- vs extratesticular; both hemiscrota plus groin; angiography for nonpalpable testis

KEY POINTS: TESTIS

- The seminiferous tubules contain developing germ cells.
- **Leydig cells produce testosterone; Sertoli cells support germ cell development.**
- Three arterial supplies: **testicular, deferential and cremasteric**.
- Testicular lymphatics drain to the **para-aortic and interaortocaval nodes**.
- Nerves implicated in chronic orchialgia: **perivasal, posterior periarterial/lipomatous and intracremasteric** complexes.
- **Sertoli tight junctions** form the anatomic component of the blood–testis barrier.
- **US is the primary imaging modality** for intrascrotal contents.

Concept — *The testis is built for **thermal and immunological isolation**: countercurrent exchange keeps it cool, Sertoli junctions sequester germ cells, and its retroperitoneal origin fixes its artery, vein, lymph and nerves to the great vessels — every surgical rule in this section protects one of these arrangements.*

EPIDIDYMIS

Anatomy

Gross Structure

- Attached to the **superior posterolateral testis**; covered by a thin layer of tunica vaginalis.
- Three portions: **caput (head), corpus (body), cauda (tail)**.
- **8–12 efferent ducts** from the rete converge in the caput into **one tightly coiled duct about 0.1 mm in diameter** (the figure legend gives 12–20 efferent ductules).
- **Caput visibly darker** — tightly compacted, narrow efferent ducts; **corpus lighter** — wider tubule. Thin septa separate the efferent-duct region from the tubule.
- **Distal tubule = sperm reservoir**; thickens as it becomes the vas, surrounded by smooth muscle.

EMBRYOLOGICAL ORIGIN — MESONEPHRIC VERSUS PARAMESONEPHRIC

Structure	Origin
Efferent ductules	Mesonephric tubules
Epididymis, vas deferens, ejaculatory duct, seminal vesicle	Mesonephric (wolffian) duct
Appendix epididymis (cyst on upper caput)	Mesonephric duct
Appendix testis	Paramesonephric (müllerian) duct
Prostatic utricle	Paramesonephric (müllerian) remnant

- The trap: the two appendages sit side by side but have **opposite origins** — appendix **epididymis wolffian**, appendix **testis müllerian**.

Microanatomic Architecture

- **Pseudostratified epithelium** throughout.

EPIDIDYMAL EPITHELIAL CELL TYPES

Cell	Distribution	Function
Principal	Throughout	Transiently ciliated; main source of extracellular vesicles regulating the luminal microenvironment; tight junctions form the blood-epididymis barrier
Basal	Base of epithelium, entire length	Proposed endocrine and stem cell roles
Narrow	Initial segment (some species)	—
Clear	Caput, corpus, cauda	Luminal pH regulation (acidification)

- **Efferent ducts**: many **ciliated cells**, irregular lumen, thinner epithelium; **cilia absent beyond the efferent ducts**.
- Cauda **less developed in humans**; transit time variable; human distal epididymis is a **weaker reservoir** than in other mammals.
- Acidic specialised microenvironment → **sperm maturation and storage in a quiescent state** before ejaculation; acidification controlled differently by each cell type.
- Epithelium rich in **oestrogen receptors** — possible **water reabsorption**.
- **Segment-specific gene and protein expression** — increasingly relevant to sperm epigenetics and fertility.

Arterial Supply

- **Testicular artery branch** → **head and body**, dividing into superior and inferior epididymal branches.
- **Deferential artery** → **tail**; **cremasteric artery** also contributes.
- **Deferential and cremasteric arteries compensate** after testicular artery ligation.
- Arteries enter through the **connective tissue septa**; coiled vessels straighten into the microvascular bed.

Venous Drainage

- Directly into the **pampiniform plexus**, or via veins communicating with the **vasal veins**; the **vasal-caudal anastomosis** is important for collateral outflow of the whole plexus.

Lymphatic Drainage**EPIDIDYMAL LYMPHATICS — CAPUT AND CORPUS VERSUS CAUDA**

Caput and corpus	Cauda
With internal spermatic vein → preaortic nodes (as for testis)	With lymphatics of the vas → external iliac nodes

Nerve Supply

- **Intermediate and inferior spermatic nerves** from the **hypogastric and pelvic plexuses**.
- Sympathetic fibres **sparse proximally** (caput, efferent ducts), forming a peritubular plexus near vessels.
- Nerve density **rises progressively from the mid-corpus to the cauda**, matched by **increasing smooth muscle**.

Imaging

Ultrasonography

- **Adjunct to examination** for equivocal findings; **highly sensitive for cysts and dilation**.
- Normally **homogeneous and isoechoic with the testis**; body and tail may be hyper- or hypoechoic.
- In a small minority the epididymis is **inverted** — **caput at the inferior pole**.
- Mean diameters: **head 9 mm, body 3.5 mm, tail 4.8 mm**; appendix epididymis seen in a minority, about 2 mm long.
- **Head size associates with normal sperm morphology**; colour Doppler detects flow in all regions; veins may extend to either testicular pole.

Magnetic Resonance Imaging

- **Multiparametric MRI**: high soft-tissue contrast, functional data, contrast-enhancement sensitivity; **less operator-dependent** → better standardisation.
- Characterises benign and malignant masses; growing role in **male infertility**.
- **T1 signal intensity correlates with epididymal obstruction** → predicts need for **vasoepididymostomy** rather than **vasovasostomy**.
- **Functional MRI** may detect early spermatogenic impairment and predict **sperm extraction success** in **nonobstructive azoospermia**.

KEY POINTS: EPIDIDYMIS

- The epididymis consists of the **caput, corpus and cauda**.
- Each region has a slightly different **cellular structure, gene expression and epigenetic function**.

Concept — *The epididymis is a regionally specialised duct: cell mix, gene expression and innervation change along its length to match function, from absorption and maturation proximally to storage and propulsion in the cauda — hence nerves and smooth muscle increase toward the tail.*

VAS DEFERENS

Anatomy

Gross Structure

Course: contiguous with the distal epididymis → **convoluted for the first 2–3 cm** → straightens, lying **posterior to the spermatic vessels** in the cord → inguinal canal → separates from the gonadal vessels → **pelvic side wall** → crosses **anterior to the ureter** → behind the bladder base, above the prostatic base → **ampulla** → joins the seminal vesicle → **ejaculatory duct** within the prostatic urethra.

- Total length **30–35 cm**; outer diameter **1.5–4.5 mm** in healthy fertile men; **lumen 0.2–0.7 mm** (mean 0.43 mm).
- Derives from the **mesonephric (wolffian) duct**, with the epididymis and ejaculatory duct.

Microanatomic Architecture

- Outer **adventitia** carrying vessels and small nerves.
- Thick **three-layer smooth muscle wall**: **outer longitudinal** → **middle circular** → **inner longitudinal**.
- Mucosa: **pseudostratified columnar epithelium** — columnar **principal cells with microvilli** and small hemispherical **basal cells** — on loose connective tissue stroma.

Arterial Supply

- **Deferential artery from the superior vesical artery** (the testis section also allows an inferior vesical origin).

Venous Drainage

SCROTAL VERSUS PELVIC VAS — VENOUS DRAINAGE

Scrotal vas	Pelvic vas
Deferential vein → joins epididymal veins and testicular venous arch → pampiniform plexus	Pelvic venous plexus

Lymphatic Drainage

- About **one-fifth of cord lymphatics lie around the vas**; drainage to **external and internal iliac nodes**.

Nerve Supply

- **Densely innervated** — about half of the nerves in the cord and spermatic fascia cluster around the vas.
- **Sensory fibres** may transmit pain in **chronic scrotal pain**.
- **Sympathetic fibres** (hypogastric → presacral nerve; deferential and testicular plexuses) drive **vasal peristalsis** and **vasoconstriction**.
- **Adrenergic fibres in all three muscle layers, densest in the outer longitudinal layer**; a substantial minority of fibres are **cholinergic**; fibres also contain **neuropeptide Y** and **nitric oxide**.
- Somatostatin, galanin, enkephalin and vasoactive intestinal peptide present; function unclear.

Imaging

Vasogram

Sequence: partial vasotomy at the **junction of straight and convoluted vas** → **24-gauge angiocatheter or ureteric catheter** advanced into the lumen → inject → observe.

VASOGRAM AGENT — INDIGO CARMINE VERSUS CONTRAST

Indigo carmine	Contrast
Observed endoscopically at the ejaculatory ducts	Observed on fluoroscopy

- Risk of **epithelial disruption and iatrogenic obstruction** → vasography is performed **only at the time of a concurrent reconstruction**, to localise obstruction.

Ultrasonography

- Vas appears as an **anechoic tube with two bright central lines** (anterior and posterior luminal surfaces).
- **Reliably identified in the straight suprascrotal portion** → US can **confirm congenital absence of the vas** when examination is equivocal.
- **Vasal diameter associates with progressive motility** — possible role in interpreting abnormal semen parameters.

Computed Tomography

- Not a primary modality; may incidentally show **vasal calcification**, suggesting comorbidity such as **diabetes**.

KEY POINTS: VAS DEFERENS

- The proximal vas is **convoluted, then straightens** as it travels into the scrotum and pelvis.
- The vas has a **muscular wall** and is important for **sperm transit**.
- The **lumen measures 0.2–0.7 mm**.
- The **superior vesical artery** gives rise to the **deferential artery**, which supplies the vas and is important for **collateral testicular flow**.

Concept — *The vas is a muscular, densely adrenergic conduit: its three-layer wall and sympathetic supply exist to drive emission, and the same dense innervation makes it a pain source, while its artery is the testicular collateral that vasectomy can sacrifice.*

SEMINAL VESICLES AND EJACULATORY DUCTS

Anatomy

Gross Structure

- Paired, tightly coiled tubular structures **posterior to the bladder and distal ureters, superoposterior to the prostate**.
- Each fuses with the **ampulla of the vas** → **ejaculatory duct** → prostatic urethra at the **verumontanum**.
- Ejaculatory duct orifices lie **inferolateral to the seminal colliculus at 5 and 7 o'clock**, covered by **one-way valves**.
- Embryology: **wolffian**; seminal vesicles bud from the distal mesonephric duct at **12 weeks' gestation**. Development is linked to the **ureter and kidney** → **congenital anomalies commonly coexist**.

Microanatomic Architecture

- **Retroperitoneal**; loose adventitia; thin smooth muscle in **circumferential and longitudinal** orientation.
- Muscle sheaths of seminal vesicle and vas **merge with the prostatic capsule at the base**.
- **Ejaculatory ducts: longitudinal smooth muscle only**; middle layer of vesicles and ducts collagenous.
- Mucosa: **pseudostratified columnar**; in the ejaculatory duct above a basal cell layer.

Arterial Supply

- **Vesiculodeferential artery** (from the **superior vesical**) → anterior surface near the tip.
- **Prostatovesicular branch** (internal iliac / inferior vesical) → additional supply; variants arise from the pudendal or superior vesical artery.
- **Ejaculatory duct: inferior vesical artery branches**.

Venous Drainage

- Follows the arteries: **vesiculodeferential veins and inferior vesical plexus** → **pelvic venous plexus**.

Lymphatic Drainage

- **Internal iliac nodes**.

Nerve Supply

- **Parasympathetic** from the **pelvic plexus**; **sympathetic** from the **hypogastric and superior lumbar nerves**; pelvic plexus also supplies the ejaculatory ducts.
- **Noradrenergic density exceeds cholinergic** — appropriate for smooth muscle contraction during **emission**.

- Axons also contain acetylcholine, neuropeptide Y and nitric oxide (as in the vas); parasympathetic role unclear, possibly **secretomotor**.

Seminal Fluid

- Seminal vesicles supply **about 70% of semen volume**.
- Fluid is **alkaline**; contains **fructose** (sperm energy source), prostaglandins, enzymes and **semenogelin**.
- **Semenogelin forms a gel matrix** around sperm, **preventing immediate capacitation**; the gel is **liquefied by prostate-specific antigen**.
- Seminal fluid proteomics: emerging diagnostic tool for fertility and oncology.

Imaging

Transrectal Ultrasound

- Transrectal ultrasound (TRUS) is **accurate** for echogenicity, homogeneity, thickened septa, calcification and cysts.
- The vesicle **shrinks after ejaculation**; post-ejaculation **anteroposterior diameter above 16 mm and volume above 6 mL suggest dilation or obstruction**.
- Vasal ampulla also assessed (mean about 4.4 mm).
- Vesicle volume and ampullary diameter **rise with prostate volume** → a large prostate may **compress the ejaculatory ducts**.
- **Vesicle aspiration or vesiculography** with TRUS → sperm cryopreservation and assisted reproduction, or before endoscopic treatment of **ejaculatory duct obstruction**.

Computed Tomography

- Vesicles show soft-tissue or fluid attenuation; CT best for **stones and congenital anomalies**; MRI better for noncongenital disease.

Magnetic Resonance Imaging

- **MRI is the modality of choice** (high spatial resolution, multiplanar). Seminal fluid is **T2 hyperintense** within a cystic-appearing vesicle with thin **intermediate-to-low T2 septa**.

Endoscopy

- Ejaculatory duct orifices at the **distal lateral verumontanum, 5 and 7 o'clock**; **utricle orifice midline at the distal-most verumontanum**.
- **Seminal vesiculoscopy** through the native ducts, with or without **laser fenestration** or **transurethral resection of the utricle**.
- Instrument: **4.5–6.9 Fr paediatric flexible ureteroscope**; an **adjunct to MRI** before planned endoscopic intervention.

SEMINAL VESICLE IMAGING COMPARED

Modality	Best for
TRUS	Size before/after ejaculation, obstruction, cysts; guides aspiration and vesiculography
CT	Stones, congenital anomalies
MRI	Modality of choice ; noncongenital disease
Vesiculoscopy	Direct inspection before endoscopic treatment

KEY POINTS: SEMINAL VESICLES AND EJACULATORY DUCTS

- The unobstructed seminal vesicle is **less than 16 mm in diameter and less than 6 mL in volume**.

- The vesicles have **sympathetic and parasympathetic** innervation and promote sperm function and ejaculation.
- The ejaculatory ducts form from the vesicles and ampullae of the vas bilaterally and empty into the prostatic urethra at the **verumontanum**.

Concept — *The seminal vesicle is an emission reservoir under noradrenergic control; a normal vesicle empties with ejaculation, so a vesicle that stays large after ejaculation is the imaging signature of distal obstruction.*

PROSTATE

Anatomy

Gross Structure

- Lies just inferior to the bladder; **base at the bladder neck, apex distal-most**. The prostatic urethra **narrows toward the apex** at the striated external sphincter (urogenital diaphragm).
- **Internal urethral sphincter**: bladder neck → through the prostate → external sphincter; longitudinal and circular smooth muscle; circular fibres **thickest at the bladder neck, thinnest at the membranous urethra**; key to **continence at rest**.

ZONAL ANATOMY OF THE PROSTATE (MCNEAL)

Zone	Share of glandular tissue	Location	Significance
Peripheral	70%	Apical, posterior and lateral bulk	Most prostate cancers ; relatively hyperechoic on TRUS
Central	25%	Surrounds the ejaculatory ducts , projects under the bladder base; ducts follow ejaculatory duct trajectory	—
Transition	Remainder	Surrounds the urethra proximal to the ejaculatory ducts	Fastest growth with age; benign hyperplasia → outlet obstruction ; heterogeneous on TRUS
Anterior fibromuscular stroma	Nonglandular	Bladder neck to striated sphincter	—

- What the table tests: **zone predicts disease** — cancer peripheral, hyperplasia transitional; the central zone is defined by the ejaculatory ducts.

Prostatic capsule

- Capsule-like layer **0.5–2 mm thick**, varying with smooth muscle content; collagen, smooth muscle, small vessels and nerves, lined with adipose tissue.
- **True capsule versus extension of parenchyma** — **controversial**.
- Best developed **posteriorly and laterally**; at the **base blends with outer longitudinal detrusor**; at the **apex glands blend with the striated sphincter**; anterolaterally blends with **endopelvic fascia**.

Fascia and support

- **Puboprostatic ligaments** suspend the prostate from the pubis near the apex.
- Parietal and visceral endopelvic fascia meet at the **arcus tendineus fascia pelvis**.
- **Cavernous nerves run within the parietal pelvic (lateral prostatic) fascia** → preserve this tissue at **radical prostatectomy** for erectile function.

- Below the arcus the capsule separates from the endopelvic fascia; the gap holds **fatty areolar tissue and lateral branches of the dorsal venous complex**.
- The **superficial branch of the dorsal venous complex** lies in the **retropubic fat anterior to the prostate, outside the prostatic fascia**.
- **Pubococcygeus** (levator ani) lies close to the endopelvic fascia along the lateral prostate.

Figure legends

- Fascial layers in cross section (Walz, 2007): **lateral prostatic fascia, endopelvic fascia, levator ani fascia, Denonvilliers fascia** (anterior and posterior laminae), **neurovascular bundle** posterolaterally between prostate, levator and rectum, with lateral nerves.

Embryology and size

- Forms under **androgen control** by masculinisation of the **urogenital sinus**: epithelial buds → lumens → branching tubules with alveoli in smooth muscle stroma.
- Female homologue: **paraurethral (Skene) glands**.
- Normal prostate **about 25 mL, 3 × 3 × 5 cm; enlarges with age, fastest in the transition zone**.

Microanatomic Architecture

- Stroma: **smooth muscle within a collagen scaffold**, rich in myosin, desmin and α -actin.
- Glands contain **basal, luminal and neuroendocrine cells**; ducts and acini have basal epithelium; lumen lined by **tall columnar secretory cells with microvilli**.
- Secretion path: acini → minor then major ducts → urethra; each acinus wrapped in connective tissue and smooth muscle; **interstitial cells of Cajal** may aid transport.
- **Neuroendocrine cells** regulate exocrine secretion and growth; malignant change → **primary neuroendocrine tumours**.
- **Urothelium** extends into prostatic ducts at their urethral junction in the transition zone.
- Internal sphincter surrounds the small periurethral glands with longitudinal smooth muscle.
- **Benign hyperplasia in the transition zone → bladder outlet obstruction**.
- **Microacinar proliferation with loss of the basal cell layer = malignancy**, mostly peripheral zone.

Ejaculatory Fluid

- Prostatic secretion is **rich in zinc and citrate**, with proteins and enzymes important for fertility.
- **Prostate-specific antigen (PSA)**: cleaves **insulin-like growth factor binding protein-3** (possible local growth promotion); expression from tall columnar cells is **androgen-dependent**.
- **PSA normally does not cross the epithelial basement membrane**; disrupted architecture (**inflammation, malignancy**) lets it leak into stroma and circulation → **raised serum PSA**.

Arterial Supply

- Internal iliac → anterior division → superior and inferior vesical, obturator, internal pudendal arteries (among others).
- Most of the **inferior vesical artery runs posterolateral to the prostate** as part of the **neurovascular bundles with the cavernous nerves**, ending at the pelvic diaphragm.
- **Prostatic artery origin is variable**: terminal inferior vesical, anterior division directly, obturator or internal pudendal; rarer origins exist.

PROSTATIC PEDICLES — SUPERIOR VERSUS INFERIOR

	Superior pedicle	Inferior pedicle
Status	Main supply ; medial (central gland) branch	Lateral (capsular) branch
Territory	Superior prostate, central zone; medial branch → median lobe and proximal two-thirds of prostatic urethra; lateral branch → lateral lobes and distal third	Apex and peripheral zone
Connection	Lateral branch of superior pedicle anastomoses with the inferior pedicle ; variable branching forms an intraprostatic network	

- Extraprostatic anastomoses with **internal pudendal, rectal and vesical arteries** — relevant to **resection and embolisation**.

Figure legends

- Arterial supply after Flocks (1937): the prostatic artery divides into a **urethral group** and a **capsular group**, with minor branches from the middle rectal (haemorrhoidal) and pudendal arteries.

Venous Drainage

- Abundant **periprostatic plexus**, which joins the **deep dorsal vein of the penis** as the **dorsal venous complex** — **ligated at radical prostatectomy**.
- Complex divides into **anterior and posterior divisions**; veins run in and branch from the neurovascular bundles posterolaterally, entering the ventral venous plexus.
- Final drainage: **internal iliac (hypogastric) veins**.

Lymphatic Drainage

- **Primary sites: obturator and internal iliac nodes**; additional **presacral** drainage.

THREE ROUTES OF PROSTATIC LYMPH DRAINAGE

Route	Destination
Superolateral	Pelvic side wall — internal and external iliac
Posterior	Presacral nodes directly
Apical	Along the internal pudendal vessels

- Sentinel node studies show **variable patterns** — implications for the extent of **lymph node dissection** at radical prostatectomy.

Nerve Supply**PARASYMPATHETIC VERSUS SYMPATHETIC PROSTATIC INNERVATION**

	Parasympathetic	Sympathetic
Source	Pelvic nerve , sacral cord	Lumbar cord → hypogastric nerve , synapsing near the organ
Transmitter	Cholinergic	Adrenergic, along capsule and periductal smooth muscle
Function	Glandular secretion	Expulsion of secretions during emission via α1 (and β2) receptors

- Nitric oxide, neuropeptide Y and vasoactive intestinal polypeptide may modulate secretion and contraction; roles unclear.
- **Bilateral afferents** to lumbar dorsal root ganglia; afferent feedback may regulate **growth and glandular architecture**.

Imaging**Transrectal Ultrasonography**

- Grey-scale TRUS: historically the prostate modality — **no contrast, office-based, urologist-performed**.

- **Volume in mL (cm³) equals weight in grams.**
- **Ellipsoid formula: length × width × height × 0.52** ($0.52 = \pi/6$) — **underestimates** volume; an updated coefficient of **0.66** has been proposed.
- Uses: volume of prostate and vesicles, nodularity, calcification, cysts, **templated biopsy guidance**.

Technique: lateral decubitus or dorsal lithotomy → lubricated endorectal **6–8 MHz** probe inserted gently → **minimal pressure** to avoid distortion → **sagittal and transverse** imaging → **concurrent digital rectal examination** to document peripheral-zone nodularity.

- Normal appearance: homogeneous; **peripheral zone relatively hyperechoic**, central gland relatively **hypoechoic**, transition zone more heterogeneous; urethra **curvilinear on sagittal view**.
- Adjuncts that raise sensitivity: **colour Doppler, contrast-enhanced US, elastography**.
- **Transabdominal US:** rough volume estimate, mainly for glands **under 30 g**; **less accurate than TRUS**.

Magnetic Resonance Imaging

- **Multiparametric MRI (mpMRI):** a preferred modality combining **anatomic sequences** (T1-, T2-weighted) with **functional sequences** (dynamic contrast-enhanced, diffusion-weighted).
- Standardised by the **Prostate Imaging Reporting and Data System (PI-RADS): version 2 in 2016** (American College of Radiology with European Society of Urogenital Radiology), **version 2.1** more recently.

COMPONENTS OF AN ADEQUATE MPMRI STUDY (PI-RADS V2.1)

Sequence	Requirement
T2-weighted	Axial plus sagittal or coronal
Diffusion-weighted	Low, intermediate and high b-values → apparent diffusion coefficient map
Dynamic contrast-enhanced	3D (preferred over 2D) T1 gradient echo, temporal resolution ≤15 seconds

- mpMRI clearly shows the zones and accurately identifies benign and **clinically significant malignant** change.
- **1.5 T or 3.0 T:** 3.0 T may improve images but is **not proven superior**; consider an **endorectal coil at 1.5 T**.
- Suspicious lesions: US-guided biopsy by **cognitive** targeting or **US-MRI fusion** software.
- **Targeted biopsy** may reduce the need for templated TRUS biopsy and **reduce detection of clinically insignificant cancer**.

KEY POINTS: PROSTATE

- Four zones: glandular tissue in the **peripheral (70%)**, **transition** and **central** zones; the anterior zone is mostly fibromuscular.
- Most prostate cancers arise in the **peripheral zone**.
- Normal prostate about **25 mL**; enlarges with age, fastest in the **transition zone**, which may cause outlet obstruction.
- The **internal urethral sphincter** surrounds the prostatic urethra; prostatic urethra and bladder neck contribute to continence.
- Arterial supply is **variable** and forms a complex anastomotic network.
- **Obturator and internal iliac nodes** are the primary lymphatic sites, with presacral drainage in addition.
- **mpMRI** is an increasingly sensitive way to image the prostate.

Concept — *Prostatic anatomy is organised by zone and by the neurovascular bundle: zone predicts which disease arises where, and the lateral fascial plane carrying the cavernous nerves dictates the dissection that preserves erection.*

URETHRA

Anatomy

Gross Structure

- Tubular structure within the **corpus spongiosum and glans**; normal **diameter 8–9 mm**.
- Female urethra: **2–4 cm**, opening directly into the vestibule.

POSTERIOR VERSUS ANTERIOR URETHRA

	Posterior urethra	Anterior urethra
Extent	Distal to the bladder neck → perineal membrane	Perineal membrane → meatus
Segments	Prostatic, membranous	Bulbar, penile, fossa navicularis

URETHRAL SEGMENTS

Segment	Length / extent	Features
Prostatic	Length of prostate	Near the anterior prostatic surface; urethral crest ; verumontanum , utricle , ejaculatory ducts; glands open into prostatic sinuses
Membranous	2–2.5 cm , prostatic apex → perineal membrane	Thin smooth muscle plus striated sphincter
Bulbar	Just distal to membranous → suspensory ligament	Lumen widens (bulb); Cowper glands at 3 and 9 o'clock
Penile (pendulous, spongy)	About 15 cm , suspensory ligament → meatus	Dorsal within the spongiosum; uniform calibre; glands of Littre
Fossa navicularis	Glanular urethra	Dilated, narrowing at the meatus; lacuna magna (Morgagni) on the roof

Prostatic urethra

- **Urethral crest** projects from the posterior midline along the full prostatic urethra and **stops at the striated sphincter**; all glands drain into **prostatic sinuses on either side of the crest**.
- An **angle at the midpoint** (0–90°, variable) turns the urethra anteriorly and divides it into **preprostatic (proximal)** and **prostatic (distal)** segments — **functionally distinct**; all glandular elements open **distal to the angle**.
- **Verumontanum** = widening and protrusion of the crest from the posterior wall.
- **Prostatic utricle** (müllerian remnant) — a **slit at the apex of the verumontanum**; orifice cystoscopically visible, about 6 mm; the two small **ejaculatory duct openings flank it**.

Membranous urethra

- Thin smooth muscle layer, then an outer layer of **circular striated muscle, horseshoe-shaped** on the anterior surface near the apex.
- Striated muscle extends from bladder base and anterior prostate along the whole membranous urethra; **signet-ring shaped**, broad-based, narrowing through the **urogenital hiatus** of the levator to the apex.
- **Posterior portion inserts into the perineal body** throughout its length.

Bulbomembranous, bulbar and penile urethra

- **Bulbomembranous urethra**: the 2 cm within the urogenital diaphragm and striated sphincter plus the first few centimetres of bulbar urethra.
- Bulbar urethra begins a few centimetres distal to the membranous urethra; lumen widens as the **urethral bulb**; **Cowper (bulbourethral) glands empty at 3 and 9 o'clock**.

- Penile urethra positioned **more dorsally than ventrally** within the spongiosum; calibre constant except at the bulb.
- **Glands of Littre**: small mucus-secreting glands lubricating the urethra before ejaculation; open on the posterior wall of the penile urethra.
- **Fossa navicularis**: multiple pockets on its dorsal and lateral surfaces; **lacuna magna (Morgagni)** opens on the roof.

Microanatomic Architecture

- Epithelium sequence: **urothelium (prostatic) → pseudostratified columnar (membranous and more distal) → stratified squamous in the glans, keratinised near the meatus.**
- Submucosa: smooth muscle, connective and elastic tissue.
- **Preprostatic sphincter** = thickened circular smooth muscle in the proximal segment, **synonymous with the internal urethral sphincter.**
- Membranous urethra: vascular submucosa surrounded by longitudinal and circular smooth muscle = **intrinsic component of the external sphincter.**
- Glanular squamous epithelium sits on loose connective tissue over spongy smooth muscle; **no muscularis mucosae.**

Arterial Supply

- **Internal pudendal artery → bulbourethral branches → urethra, corpus spongiosum and glans.**

Venous Drainage

- **Pudendal or prostatic plexuses → internal pudendal vein.**

Lymphatic Drainage

- Grossly: **internal iliac (hypogastric) and common iliac nodes.**
- **Membranous**: runs anterior to the prostate, joins anteroinferior bladder lymphatics → **anterior or medial retrofemoral nodes and middle node of the medial external iliac group.**
- **Penile**: mucosal network, longitudinal channels with transverse and oblique anastomoses → trunks at the **bulbomembranous urethra.**
- **Bulbomembranous**: variable — along the urethral artery or artery of the bulb, or behind the symphysis to the medial retrofemoral node.

Nerve Supply

STRIATED VERSUS INTRINSIC SMOOTH MUSCLE SPHINCTER

	Striated sphincter	Intrinsic smooth muscle
Type	Somatic motor	Autonomic
Nerves	Pudendal nerve; plus a somatic sacral-plexus branch on the levator ani surface	Cavernous nerves (from the pelvic plexus)
Segments	Mainly ventral root S3 , with S2 contribution	—

- Sensory fibres from the striated sphincter travel in the **pudendal nerve, mainly S2** (lesser S3); the sphincter's somatic motor neurons lie in **Onuf nucleus.**
- **Penile urethra**: sensation via submucosal axons that run centrally in the **dorsal nerve of the penis.**

Imaging

Ultrasonography

- **Sonourethrography**: noninvasive, **best for the anterior urethra**; mainly **stricture identification and characterisation**; also calculi, diverticula, abscesses, foreign bodies.

Magnetic Resonance Imaging

- **Best cross-sectional modality, no segmental limitation**; urethral cancer, periurethral masses, involvement by adjacent disease; especially helpful in **urethral malignancy**.

Endoscopy

- Direct visualisation of the **entire lumen**; flexible or rigid cystoscopes, **6–25 Fr** depending on the scenario.
- Endoscopic sequence: bladder neck → prostatic (verumontanum) → membranous → bulbar → fossa navicularis.

Fluoroscopy

- Common for intraluminal processes: **anatomy, stricture, trauma, postsurgical change**; contrast **antegrade or retrograde**.
- Urethrography positioning: **slight lateral decubitus** for the most accurate representation.
- Retrograde urethrogram landmarks: prostatic urethra → verumontanum (ejaculatory ducts enter) → **membranous urethra with physiological narrowing from the external striated sphincter** → bulbar → pendulous urethra.

URETHRAL IMAGING COMPARED

Modality	Scope	Main use
Sonourethrography	Anterior urethra	Stricture; calculi, diverticula, abscess, foreign body
MRI	All segments	Malignancy, periurethral masses
Endoscopy	Entire lumen	Direct inspection
Retrograde/antegrade urethrography	Entire lumen	Stricture, trauma, postsurgical anatomy

KEY POINTS: URETHRA

- The urethra lies within the **corpus spongiosum and glans**.
- The **anterior urethra** runs from the perineal membrane to the meatus.
- The **posterior urethra** begins distal to the bladder neck; the transition to anterior urethra is at the **perineal membrane**.
- The epithelium changes to **squamous where the urethra traverses the glans at the fossa navicularis**.
- Arterial supply: **internal pudendal artery**, whose bulbourethral branches supply urethra, spongiosum and glans.
- The **verumontanum** is the widened, protruding urethral crest on the posterior wall.
- The **prostatic utricle** (müllerian remnant) orifice is a slit at the apex of the verumontanum.

Concept — Each urethral segment is named for what surrounds it — prostate, sphincter, bulb, spongiosum, glans — and the **perineal membrane** is the single boundary dividing posterior from anterior, with sphincteric function, epithelium and preferred imaging changing across it.

PENIS

Anatomy

Gross Structure

- **Paired corpora cavernosa** (erectile bodies) extend proximally as the **crura**, attached to the **pubic arch**.
- The urethra runs in the **corpus spongiosum**, whose proximal part is the **bulb**; the **glans** is the **distal expansion of the spongiosum**.
- **Dorsum** = superior surface in erection; **ventrum** = inferior surface, containing the urethra.
- Corpora join beneath the pubis (**penile hilum**) and form most of the shaft.
- **Septum is incomplete and permeable distally** → free vascular communication → the corpora function **physiologically and pharmacologically as a single unit**.

Tunica albuginea

- Tough, mainly **collagenous** envelope of the corpora with an **elastic-fibre lattice**; mean thickness **2–3 mm**.
- Ventrally it is modified to form a **groove** accommodating the spongiosum, where the outer layer is thin.

TUNICA ALBUGINEA — INNER VERSUS OUTER LAYER

	Inner layer	Outer layer
Orientation	Circular	Longitudinal
Function	Supports and contains cavernous tissue; gives off intracavernosal pillars (struts)	Runs glans → proximal crura → inserts on the inferior pubic ramus

- In erection both layers are **tightly stretched**; flaccid, they form an **undulating meshwork**.

Spongiosum, glans and fascia

- Spongiosum tapers distal to the bulb and runs **ventral** to the corpora; **corona** separates shaft from glans base.
- **Buck fascia** surrounds the corpora dorsally and **splits to envelop the spongiosum** ventrally; it joins the glans base at the corona.

FUNDIFORM VERSUS SUSPENSORY LIGAMENT

Fundiform ligament	Suspensory ligament
Superficial ; collagen and elastic fibres from the rectus sheath , blending with and surrounding Buck fascia	Deeper ; fibres from the pubis

- Shaft skin: very elastic; only glands are **smegma-producing glands at the corona base**; **mobile** because the dartos backing is **loosely attached to Buck fascia**.
- **Prepuce**: shaft skin folded over the glans, attaching below the corona. **Glans skin immobile** — attached to the underlying tunica albuginea.

Microanatomic Architecture

- Cavernosal structure gives **flexibility, strength and rigidity**.
- Fibrous framework of **collagen (types I and IV; less type III)**, elastin and fibroblasts traverses the corpora, forming **endothelium-lined cavernous sinuses** with smooth muscle bundles.
- **Smooth muscle** ≈ 45% of corporal volume.
- Filaments: **thin** = actin, **thick** = myosin, **intermediate** = desmin or vimentin.
- **Myoendothelial junctions** (extensions through the internal elastic lamina) link smooth muscle to endothelium, with **gap junctions** at the contacts.
- **Endothelial nitric oxide synthase** in corporal endothelium — major role in **maintenance of erection**.

Arterial Supply

- **Superficial system: external pudendal** branches of the femoral vessels → skin, running longitudinally in the **dartos**.
- **Deep system: internal pudendal** arteries bilaterally, running in **Alcock (pudendal) canal** → common penile artery → three branches.

BRANCHES OF THE COMMON PENILE ARTERY

Branch	Course	Supplies
Bulbourethral	Pierces the perineal membrane; enters spongiosum from above at its posterolateral border	Spongiosum, glans, urethra
Cavernosal	Enters the corpus at the hilum , near its centre	Straight and helicine arteries → cavernous sinuses
Dorsal	Between crus and pubis; between dorsal vein and dorsal nerve, under Buck fascia, to the glans	Dorsal corpora; cavernous and circumferential branches to urethra and spongiosum

- **Variability is common:** one cavernosal artery may supply both corpora or be absent; an **accessory pudendal artery** may supplement or replace common penile branches.

Venous Drainage

- **Superficial dorsal vein:** **outside** Buck fascia. **Deep dorsal vein:** **beneath** Buck fascia, between the dorsal arteries.
- Channels at the glans base (retrocoronal plexus) form the dorsal vein, which runs in the groove between the corpora and drains to the **preprostatic (periprostatic) plexus**.

VENOUS DRAINAGE BY TERRITORY

Territory	Pathway	Destination
Skin	Superficial dorsal vein , outside Buck fascia	—
Distal two-thirds of shaft	Sinuses → subtunical plexus → emissary veins (oblique through tunica) → circumflex veins (3–10, from spongiosum around corpora) → enter deep dorsal vein perpendicularly	Deep dorsal vein → preprostatic plexus
Proximal third	Emissary veins → 2–5 cavernous veins on the dorsomedial corpora → between bulb and crura at the hilum, receiving bulbar and crural branches	Internal pudendal veins

Lymphatic Drainage

- Penile skin → **superficial inguinal and subinguinal** nodes; shaft lymphatics **converge on the dorsum and ramify to both groins**.
- **Glans:** deep to Buck fascia dorsally → **superficial and deep inguinal nodes bilaterally**.
- Proposed direct glans-to-pelvic channels and sentinel nodes medial to the superficial epigastric vein **have been challenged**.

Nerve Supply

- **Sensory: dorsal nerves**, richly supplying the glans, running **lateral to the dorsal arteries**; small **perineal nerve** branches supply the ventrum as far as the glans.
- **Somatic: S2–S4 via the pudendal nerve** → Alcock canal → **dorsal nerve of the penis**.
- **Autonomic: cavernous nerves** carry sympathetic and parasympathetic fibres from the **pelvic plexus**, penetrating and ramifying in erectile tissue.

Figure legends

- Dorsal arrangement under Buck fascia, midline outward: **deep dorsal vein → dorsal arteries → dorsal nerves**; the subtunical space separates tunica from erectile tissue.

Imaging

Cavernosogram

- Radio-opaque contrast into a corporal body with plain films; historically for **venous-leak erectile dysfunction**; now used **occasionally at penile fracture repair**.

Ultrasonography

- **Penile Doppler** (first described in the 1970s): erectile dysfunction, **Peyronie disease**, arteriocorporal fistula and other pathology.
- High-resolution **7.5–12 MHz** real-time US with colour-pulsed Doppler → dorsal and cavernous arteries, **haemodynamic analysis after intracavernosal injection**; whole length from crura to glans tip.
- Colour duplex: flow **toward the probe red, away blue**.

PENILE DUPLEX THRESHOLDS

Parameter	Threshold	Indicates
Peak systolic velocity	< 30 cm/s	Arterial insufficiency
End-diastolic velocity	> 5 cm/s	Veno-occlusive dysfunction
Resistive index	< 0.80	Veno-occlusive dysfunction

- What the table tests: **systolic measures judge inflow; diastolic measures judge outflow**. Venous leak lets diastolic flow persist, keeping resistance low.
- Tunica albuginea and **Peyronie plaques** can be assessed.

Magnetic Resonance Imaging

- T1- plus T2-weighted images give fine detail of corpora cavernosa and spongiosum.
- Shows **congenital anomalies, prosthesis malposition, malignancy, haematoma, post-traumatic fibrosis, penile fracture, Peyronie plaques**.
- All layers identifiable at high resolution: skin, dartos, hypodermis, corporal vascularity, urethral lumen, deep arteries, superficial dorsal veins, tunica albuginea.

KEY POINTS: PENIS

- The paired erectile bodies are the **corpora cavernosa**.
- The **glans** is an expansion of the corpus spongiosum.
- A **permeable septum** lets the corporal vascular spaces communicate.
- **Superficial arterial system from the external pudendal; deep system from the internal pudendal arteries**.
- The **dorsal vein** runs between the corpora and drains to the **preprostatic plexus**.
- Shaft lymphatics converge on the dorsum and drain to **both groins**.
- Sensation via the **dorsal nerves**; somatic supply **S2–S4 via the pudendal nerve**.

Concept — *The penis is a **pressurised compartment**: a paired but functionally single cavernous body, filled through the cavernosal arteries and emptied by emissary veins crossing an inelastic tunica — so arterial failure shows as **low peak systolic velocity** and veno-occlusive failure as **persistent diastolic flow**.*